THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **The Wonderful World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student heats two beakers: one with 100 mL and one with 200 mL of water and notes the time to boil. Which statement about fairness is true? 1

(a) it is a perfectly fair test
(b) the beakers must differ in colour
(c) time is the cause
(d) the volume of water is changed along with the time, so the test has two changing things

2. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** ice cream causes sunburn
(c) sunburn causes ice cream sales **(d)** it is magic

3. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) toss a coin **(b)** compare the methods and repeat carefully
(c) ignore both **(d)** believe one group

4. In an experiment on how salt affects the boiling point, two things were changed: salt amount and heater power. The error is: 1

(a) more than one variable was changed **(b)** water is cold
(c) too many readings **(d)** salt is a solid

5. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) amount of soil **(b)** type of plant **(c)** amount of light **(d)** amount of water

6. Which of the following is not good practice while recording data? 1

(a) noting the date
(b) rounding off numbers to look neat without noting it
(c) giving a title
(d) writing units

7. A scientist is convinced that her hypothesis is right, but the data shows otherwise. The correct action is to: 1

(a) discard the data **(b)** repeat until the data matches
(c) change the readings **(d)** revise the hypothesis

8. In the above experiment, which variable is measured as the outcome? 1

(a) growth of the plant **(b)** amount of water
(c) soil **(d)** sunlight

9. Which one of these questions leads to a hypothesis most directly? 1

(a) What is the date today?
(b) Why do wet clothes dry faster on a windy day?
(c) Who is taller?
(d) What do you want to eat?

10. Which of these is the best example of a testable question? 1

(a) Does temperature affect how fast sugar dissolves?
(b) Which sport is exciting?
(c) Is a mango tastier than a pear?
(d) Is blue a better colour than red?

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In an experiment on plant growth, A is kept in sunlight and B in shade, but A is also given more water. Why is this unfair? 2

12. How can a scientific model such as a globe help us? 2

13. Give one example in which technology causes a problem and how science may solve it. 2

14. Why do scientists use a control group? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Design a fair test to compare which of three fabrics dries the fastest. Name controlled variables. 3

16. Explain the role of curiosity, observation and reasoning in the discovery of gravity by Newton. 3

17. Explain the term 'variable' and give examples of three variables in an experiment on dissolving sugar. 3

OR

In a plant experiment, plants P (light, water), Q (dark, water), R (light, no water). State what each comparison tests. 3

18. Compare the use of a thermometer with feeling with the hand to judge temperature. 3

OR

Describe two ways in which a scientific idea may be proved wrong. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student wants to know if the colour of a cup affects how fast hot water cools. Design the investigation. 5

OR

Design a survey and experiment to study the effect of music on plant growth. Name the control and variables. 5

20. Three students measured the time of 20 pendulum swings: 28 s, 30 s, 32 s. Find the average, the time for one swing, and say why 20 swings are timed. 5

OR

Explain how errors can occur in measurement and suggest five ways to reduce them. 5

END OF QUESTION PAPER — Total: 40 Marks

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